

Call Box User Manual

V1.0



Thank you sincerely for purchasing our products.
This manual describes the operation instructions
and safety precautions of the product.
Please read this manual thoroughly and pay
attention to safety during operation.
Keep this manual in a proper place for future
reference.

Ezhan

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一、Safety Instructions

1.1 Safety Precautions

The precautions specified herein are intended to ensure safe and proper use of the product and prevent potential hazards, so as to avoid injury to yourself and others. Please fully understand these contents before using this product.



Notice

- Do not use this product in environments with explosive gas, flammable gas, corrosive atmosphere, places prone to water exposure or near combustibles. Failure to comply may result in product malfunction, fire or personal injury.
- To avoid risks of fire and electric shock, operate the product under rated voltage only.
- Perform wiring strictly in accordance with the connection diagram; otherwise, fire hazard may occur.
- Always cut off the power supply before maintenance and inspection, otherwise personal injury may be caused.
- Do not disassemble or modify this product. Such actions may lead to personal injury or equipment damage. Please contact our company if internal inspection or repair is required.

二、Product Overview

- **Product Introduction:** An industrial-grade call terminal developed based on dedicated chips. It is specially designed for mobile operation scenarios of AMR robots with wiring-free deployment, suitable for emergency communication and scheduling interaction of AMR clusters in smart factories and unmanned warehouses.
- **Core Positioning:** A dedicated wireless emergency communication device matched for AMRs. It replaces traditional wired fixed call devices, solves the problems of frequent disconnection and poor anti-interference performance of ordinary wireless equipment in industrial environments, and fills the gaps of remote intercom and fault reporting for unmanned AMR operations.
- **Online Status Monitoring:** Monitor device online, offline and signal status around the clock. It can timely detect communication loss and equipment abnormalities to ensure stable and reliable AMR operation links.

- **Alarm Record Tracing:** Automatically store fault alarms and abnormal records. Background query and traceability are supported to facilitate equipment maintenance, fault review and operation management.
- **Flexible Wireless Deployment:** Adopts independent WebSocket communication. No factory network cables or telephone lines are required, adapting to large-scale, cross-regional and mobile dynamic operation scenarios of AMRs.
- **Industrial Anti-interference & Stable Performance:** Equipped with an industrial-grade main control chip, deeply optimized to resist electromagnetic interference generated by factory motors and frequency converters. It features robust hardware stability, ensuring communication free of packet loss, disconnection and lag during operation.
- **Applicable Scenarios:** Developed based on the open-source ecosystem, the AMR Call Box supports secure and reliable calling and data transmission.
Applicable Scenarios: Widely adopted for AMR sorting and handling in smart warehouses, AMR material transportation on factory production lines, unmanned intelligent manufacturing workshops, and large-scale distributed AMR cluster operations within the plant.

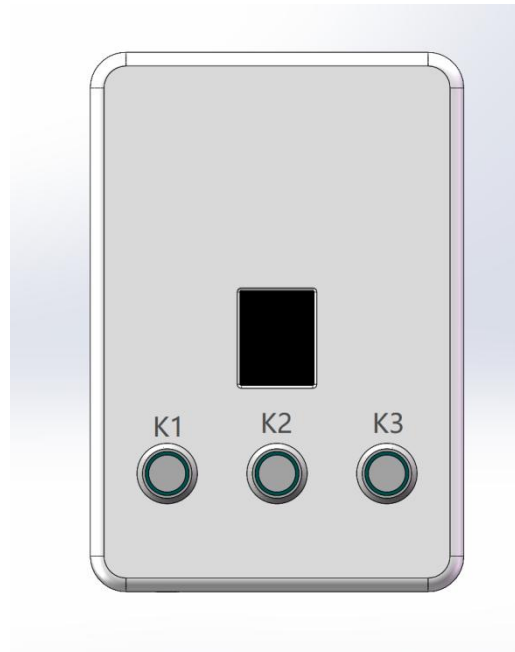
三、Product Specifications

3.1 Specification Parameters

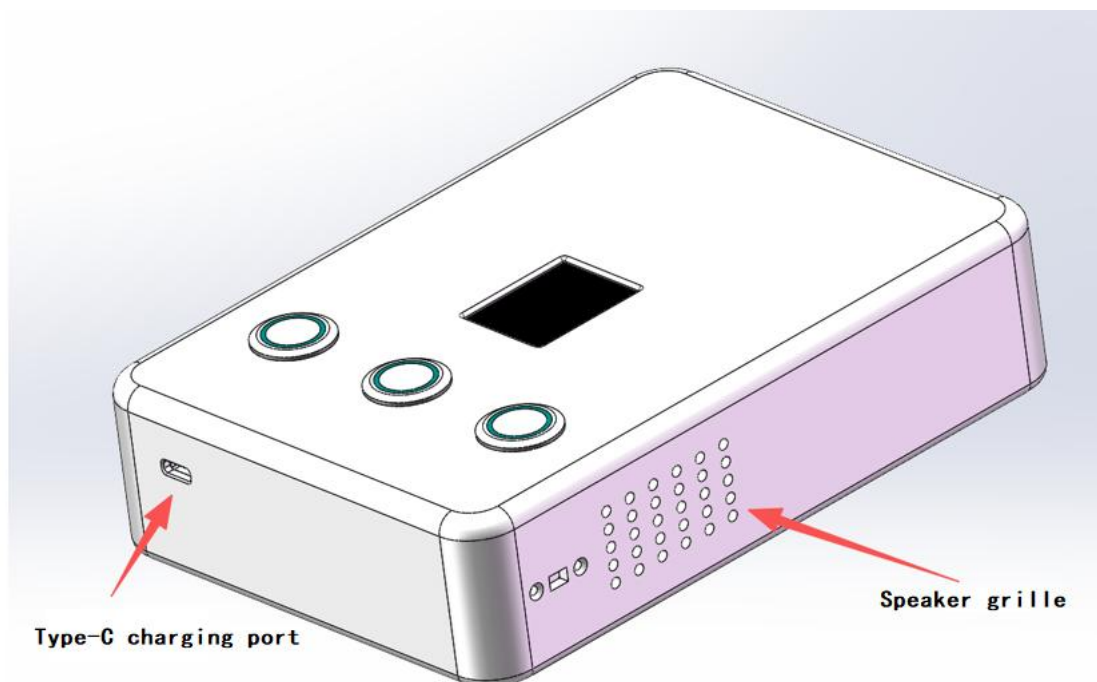
Connection Mode	Signal Connection
Transmission Distance	Communication is available within the signal coverage area.
Configuration Method	Web Page Configuration
Rated Voltage	DC5V
Number of Buttons	3 pcs
Rated Power	2W
Housing Material	Resin
Operating Temperature	-10℃~+55℃
Operating Humidity	85%RH (No condensation)
Wireless Standard	802.11 b/g/n
Wireless Frequency	2.4 GHz/5.8GHz

3.2 Product Dimensions

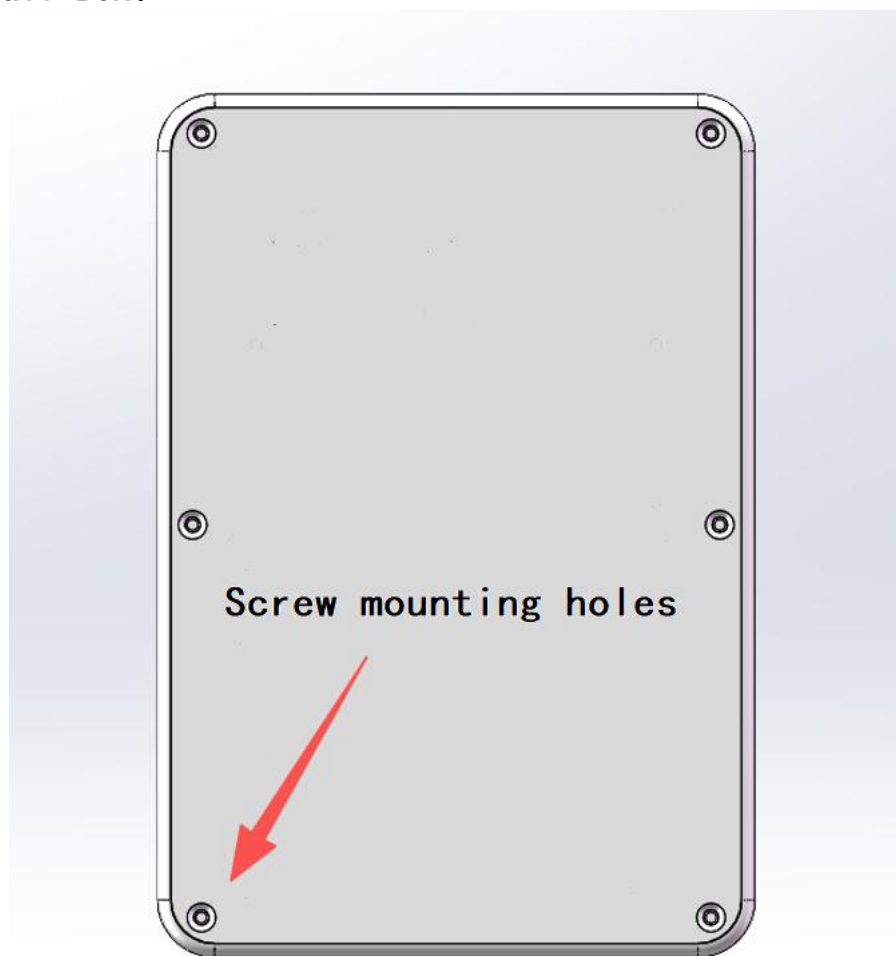
Front panel of the call box: Equipped with 3 buttons (K1, K2, K3 corresponding to Task 1, Task 2, Task 3)



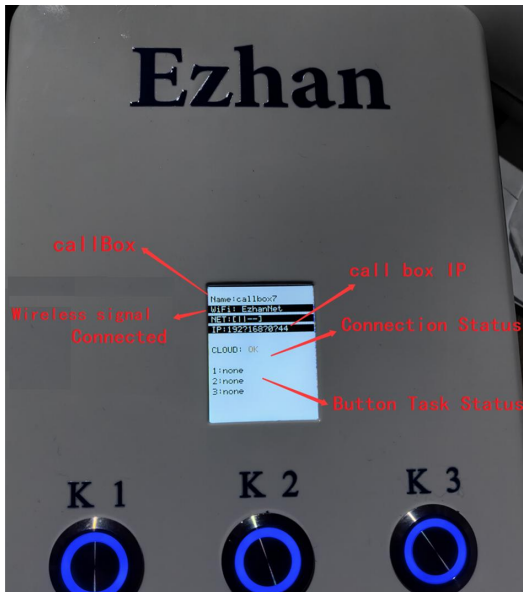
Side of Call Box:



Rear of Call Box:

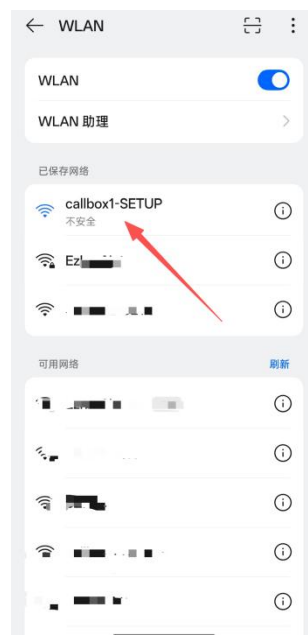


四、Call Box Parameter Configuration



4.1 Configure the Call Box

(1) Turn on the settings on your mobile phone or use a computer to connect to the signal of the call box. Find the call box name on the connection page; the name is displayed on the screen of the call box.



(2) Open the browser and enter the URL: `http://192.168.4.1/`. The following interface will appear. Save settings before proceeding to the next step after each configuration.

Callbox Setup

Setup AP: **callbox7-SETUP**
 URL: **http://192.168.4.1/**
 Name: **callbox7**
 Server IP: **192.168.0.20**
 ICCID: **0.0.0.0**
 STA: **Connected EzhanNet**

Callbox name

callbox7

Only letters, numbers and underscore, 1-16 characters.

Save name

Cloud server IP

192.168.0.17

IPv4 address, for example 8.166.134.136.

Save server IP

network

Manual input

SSID

password

Connect

4.2 Rename the Call Box

You may modify the call box name in the Callbox name field. After modification, reconnect to the network with the new name and revisit this page.

Callbox Setup

Setup AP: callbox7-SETUP
URL: <http://192.168.4.1/>
Name: callbox7
Server IP: 192.168.0.20
ICCID: 0.0.0.0
STA: Connected EzhanNet

Callbox name

callbox7 **Default name of the call box (editable)**

Only letters, numbers and underscore, 1-16 characters.

Save name

Cloud server IP

192.168.0.17

IPv4 address, for example 8.166.134.136.

Save server IP

network

Manual input

SSID

password

4.3 Configure Cloud Server Address

Fill in the IPv4 address of the robot on the same local area network as the call box in the Cloud server IP field: 120.76.142.163 (Fill in according to actual conditions).

Callbox Setup

Setup AP: callbox7-SETUP
URL: http://192.168.4.1/
Name: callbox7
Server IP: 192.168.0.20
ICCID: 0.0.0.0
STA: Connected EzhanNet

Callbox name

Only letters, numbers and underscore, 1-16 characters.

Save name

Cloud server IP

Target Host IP Address

IPv4 address, for example 8.166.134.136.

Save server IP

network

Manual input

SSID

password

Connect

4.4 Configure Network Signal of Current Call Box

In the Manual input box located at the bottom of the configuration interface, select the same network that the robot is connected to, enter the password and save the settings. ;

Setup AP: callbox7-SETUP
 URL: <http://192.168.4.1/>
 Name: callbox7
 Server IP: 192.168.0.20
 ICCID: 0.0.0.0
 STA: Connected EzhanNet

Callbox name

Only letters, numbers and underscore, 1-16 characters.

Save name

Cloud server IP

IPv4 address, for example 8.166.134.136.

Save server IP

network

Manual input

SSID

password

Connect

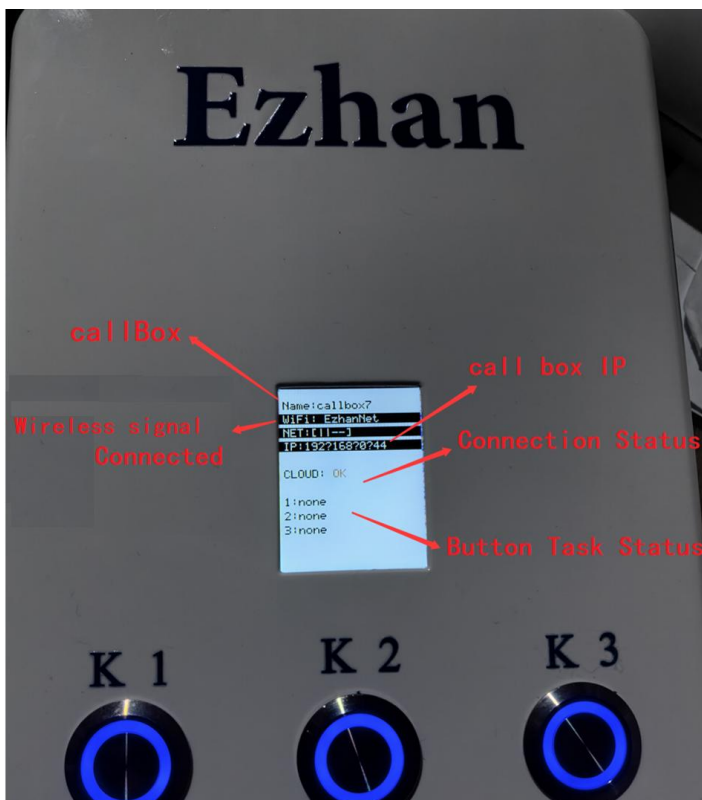
Choose the network with optimal RSSI. RSSI is negative: the nearer to 0, the stronger signal.

WiFi network

Manual input

Manual input
EzhanNet_test RSSI:-46 CH:1 WPA/WPA2
EzhanNet_test RSSI:-49 CH:40 WPA/WPA2
EzhanNetA RSSI:-63 CH:36 WPA/WPA2
ChinaNet-Hzts RSSI:-75 CH:11 WPA2
EzhanNet RSSI:-78 CH:6 WPA/WPA2
EzhanNet RSSI:-78 CH:36 WPA/WPA2
ChinaNet-Hzts-5G RSSI:-79 CH:48 WPA2
ChinaNet-Hzts RSSI:-80 CH:11 WPA2

After saving successfully, the robot IP and the local IP of the call box will be displayed on the call box, as shown in the figure below.

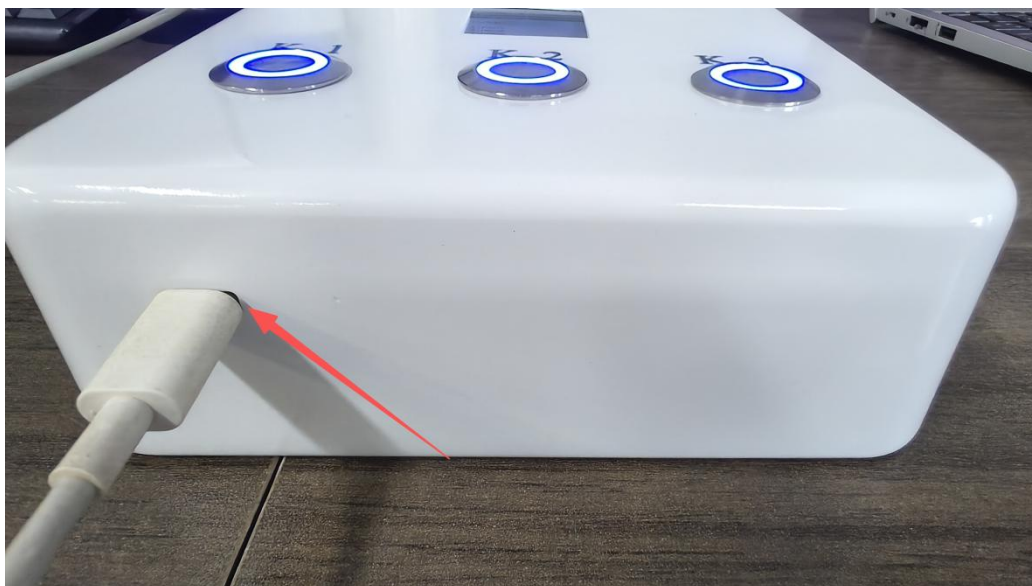


五、Power-on Operation and Call Box Management

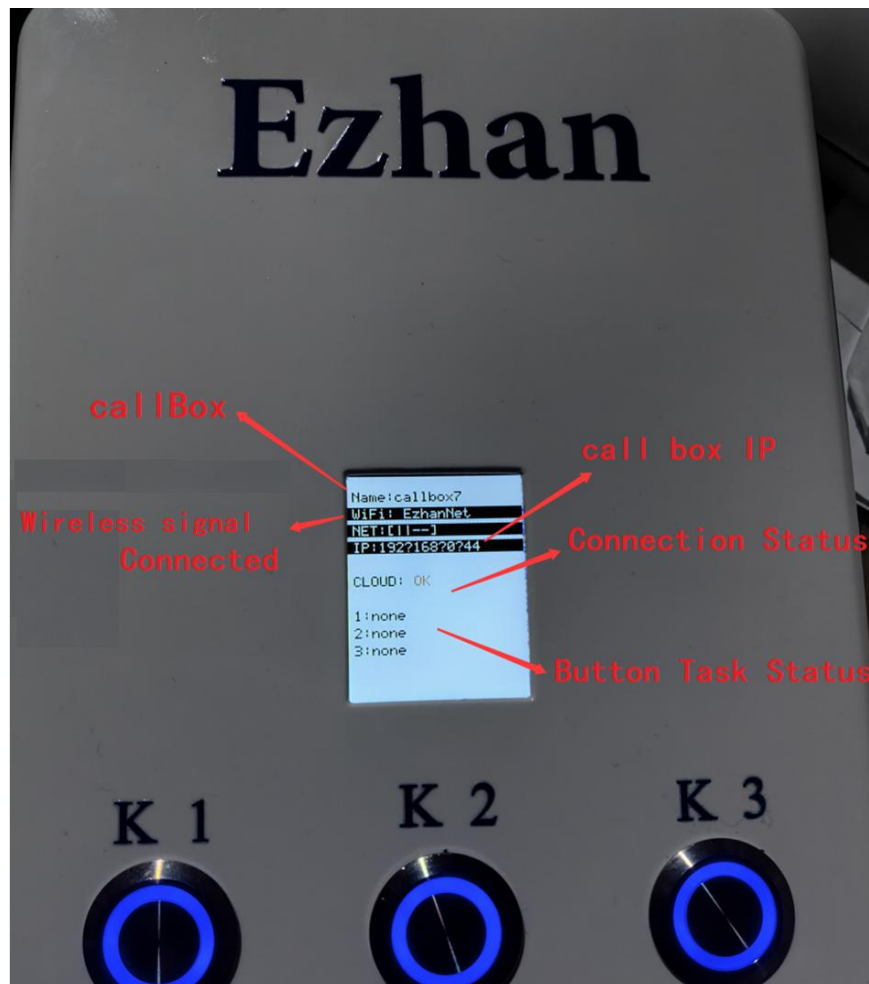
5.1 Prepare Type-C data cable



Plug the Type-C data cable into the call box interface. After power-on, the call box button indicator light and screen will turn on.

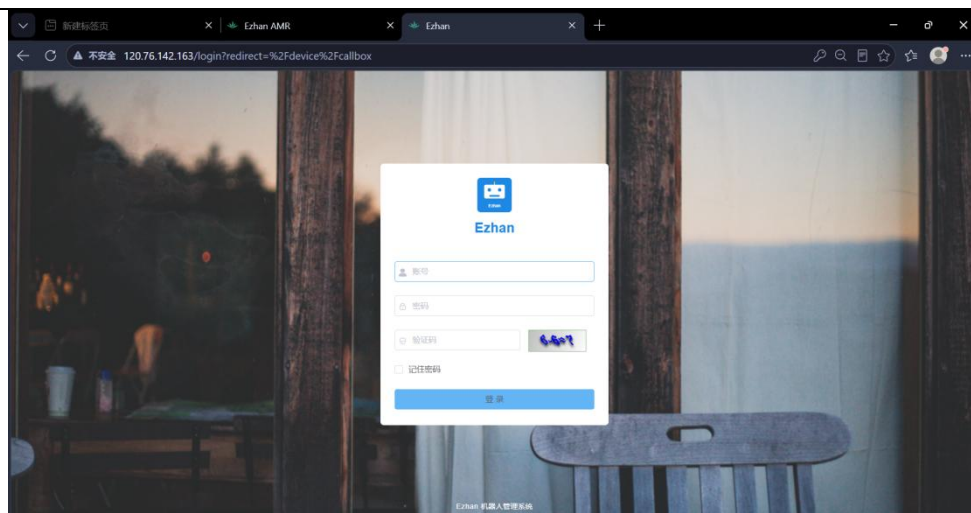


5.2 Screen Information Overview

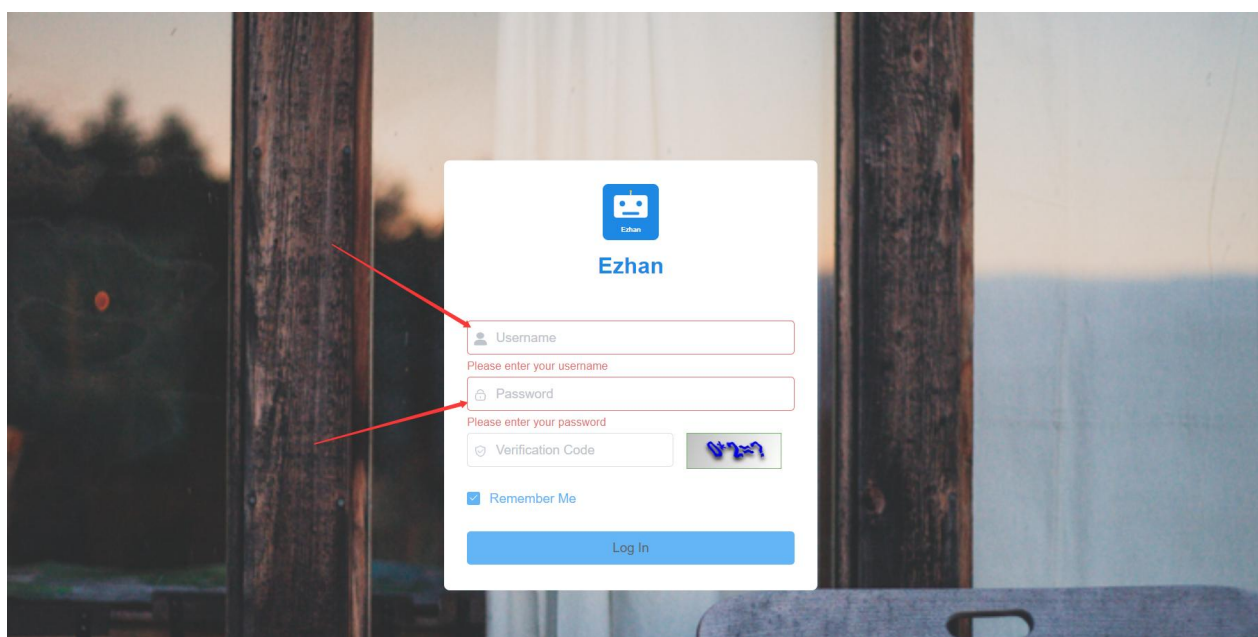


5.3 Set Task Buttons

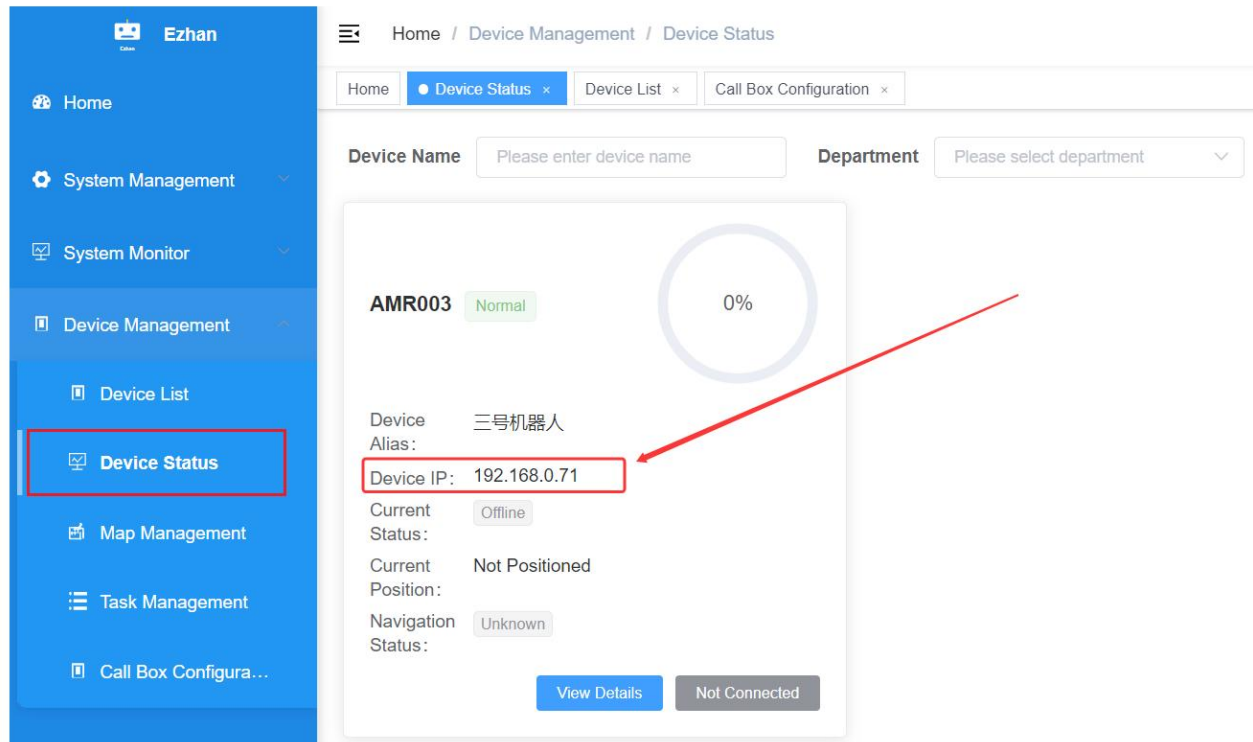
5.3.1 After running the scheduling system, enter the URL ``http://127.0.0.1/device/callbox`` in the browser (adjust according to actual conditions) to access the interface as shown below:



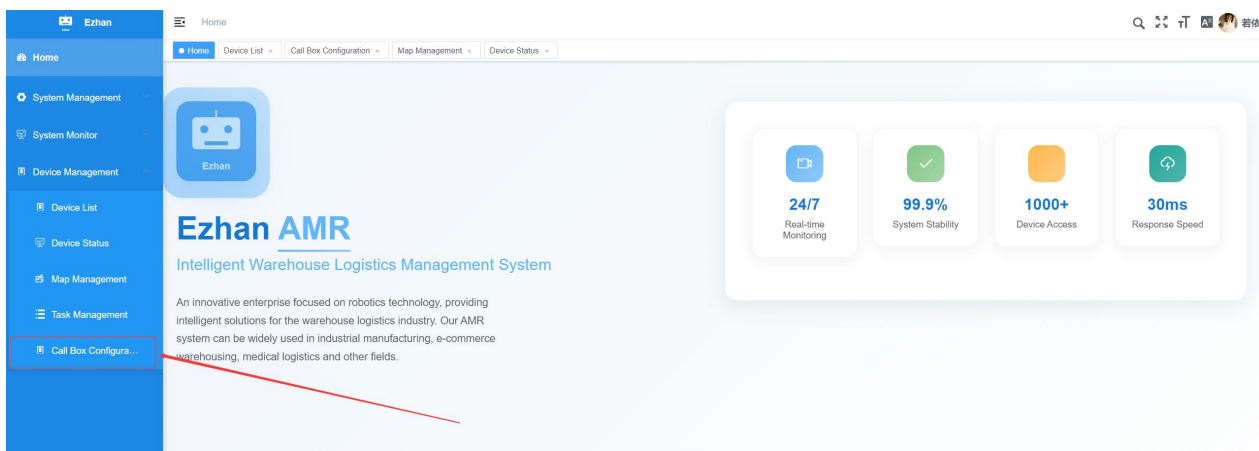
5.3.2 Enter the user account and password



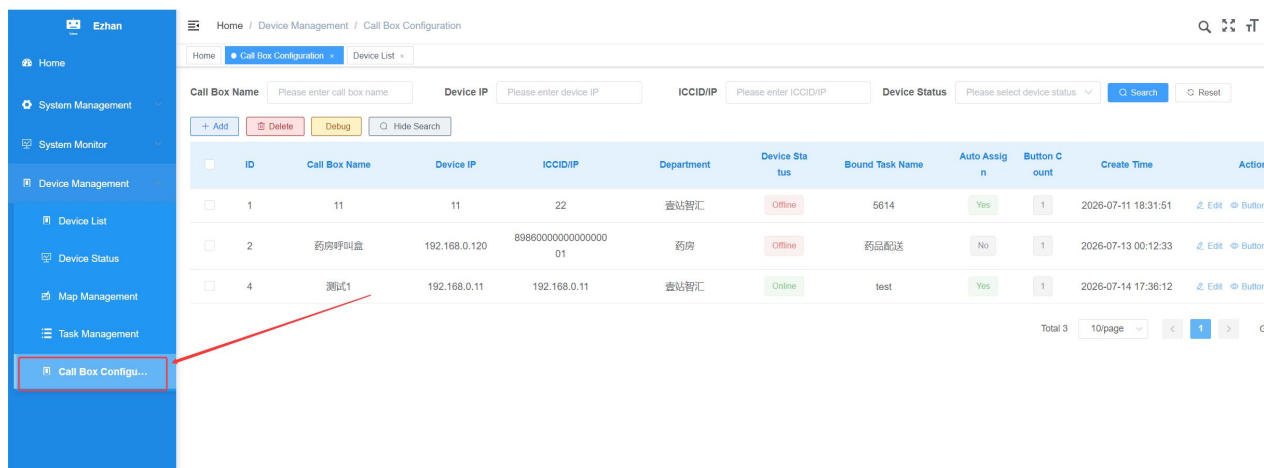
5.3.3 After entering the home page, click Device Status to obtain and copy the robot device IP address.



5.3.4 Click Configure Call Box



5.3.5 After entering the call box configuration interface, click New Call Box.



5.3.6 Fill in Call Box Information for Cloud Configuration

Enter the name of the call box to be bound in the Call Box Name field;

Input the robot device IP address obtained in Section 4.4 in the Device IP column;

Fill in the corresponding IP address of the call box under ICCID/IP;

Select Yizhan Zhihui for the department (select according to actual situation);

Click Save after completing all information.

Add Call Box Configuration

* Call Box NamePlease enter call box nameName the call box in the dispatching system. If its function is medicine delivery, it can be named Medicine Delivery Call Box

* Device IPPlease enter device IP(WAN IP address connected by the AGV trolley)

* ICCID/IPPlease enter ICCID/IP (SIM card unique identifier)call box IP

* DepartmentPlease select department

Bound Task NamePlease enter bound task name (manual edit)Task remarks for the call box: fill in Medicine Delivery if it is in charge of drug delivery.

Auto Assign☐ Disable☒ EnableWhen enabled, tasks will be automatically assigned to idle robots in the same department

After automatic assignment is enabled, if the AGV trolley is executing a task, the task issued by this call box can be terminated and reassigned.

Device Status☐ Online☒ Offline

Button name for matched call box: configurable as K1, K2, K3 or other custom text.

Button Configur...

Button ID	Button Name	Bound Task	Bound Station	Action
1	Button Na	Bound Task Name	Station name, separated by o	

+ Add ButtonAdd the corresponding number of buttons according to the physical buttons on the call box.

RemarkPlease enter remark

Confirm

Cancel

5.3.7 Button Configuration

Click the Add button, fill in the name of the button to be bound on the call box and the configured jacking task name. Up to 3 buttons can be added.

Add Call Box Configuration

* Call Box Name

* Device IP

* ICCID/IP

* Department

Bound Task Name

Auto Assign ☒ Disable ☐ Enable When enabled, tasks will be automatically assigned to idle robots in the same department

Device Status ☐ Online ☒ Offline

Button Configur...

Button ID	Button Name	Bound Task	Bound Station	Action
No Data				
+ Add Button				

Remark

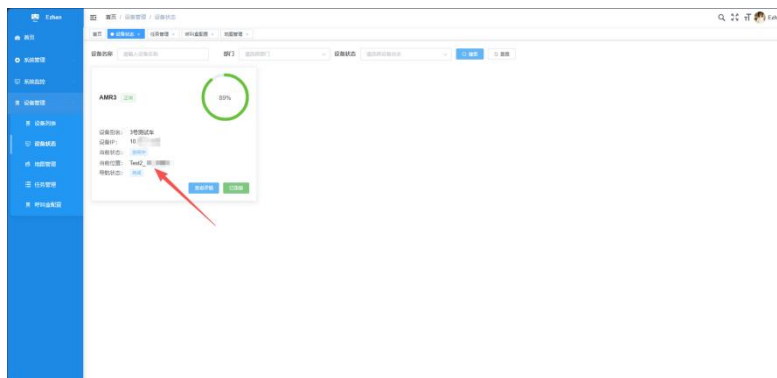
[Confirm](#) [Cancel](#)

5.3.8 Introduction and Configuration of Station Binding

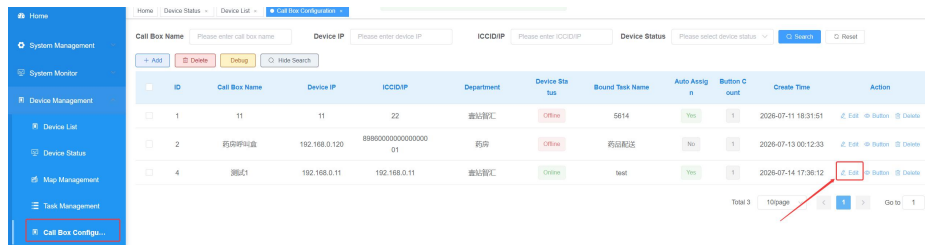
Function

The station binding function enables voice announcement when the robot arrives at the task station, providing voice reminders upon the robot's arrival.

Configuration Method: Obtain and copy the robot's current position from the device status section.



Click Edit on the call box configuration page.



Paste the copied current location at the site binding field.

Edit Call Box Configuration

Call Box Name

Device IP

ICCIDIP

Department

Bound Task Name

Auto Assign

Disable

Enable

When enabled, tasks will be automatically assigned to idle robots in the same department

Device Status

Online

Offline

Button Configuration

Button ID	Button Name	Bound Task	Bound Station	Action
1	1	jackTask	Station name, separated by 0	[Delete]

Add Button

Remark

Confirm

Cancel

六、Troubleshooting

Fault Description	Inspection Items	Corrective Actions
No response on call	Check power supply status	Ensure stable power supply for call box
	Is signal configuration OK?	Reconfigure call box according to operating steps
	Is the call box signal connected normally	Install an additional router or place the call box in an environment with good signal.

七、After-sales Service

7.1 Warranty Term

Limited warranty for call box: Free maintenance for failures caused by quality defects and design flaws within warranty term.。

Warranty Period: 1 year from the date of sale。

7.2 Warranty Coverage

Products within the warranty period that meet warranty terms shall be repaired or replaced free of charge.

Malfunction upon unpacking caused by damage in transit

Malfunction caused by damaged original components

Inoperability resulting from inherent design defects

7.3 Limitation of Liability

Please note that we will not provide free after-sales service or warranty under the following circumstances during product operation:

Product damage caused by improper installation not in accordance with the user manual;

Product damage resulting from operation under inappropriate environmental conditions;

Product damage incurred due to non-standard operation against the specifications in the user manual;

Unauthorized disassembly or maintenance of the product without the company's written approval;

Product damage caused by force majeure events including natural disasters, fire and other irresistible external factors.